

Plan d'étude parcours 3IA

	UE	ECUE	Total hours	Credits (ECTS)	P1 (Hours)	P2 (Hours)
Semestre 5	Software and data management	Software engineering	42	3,5	21	21
		Data base management system	42	3,5	21	21
	Web development	Python framework for the web	42	4	21	21
	Data engineering	Python for Data Engineering	21	2	21	0
		Data warehousing	42	3	42	0
		Machine learning basics	42	3	0	42
		Workshop "Fundamentals on Deep Learning"	18	1	18	0
	Networking and Systems	Networks and TCP/IP Protocols	42	3	21	21
Semestre 6	Estimation Techniques for Engineers	Estimation Techniques for Engineers	42	4	21	21
	Communication, Culture and Citizenship A3	Communication, Culture and Citizenship A3	42	3	21	21
	Object-oriented Programming	Object-oriented design and Java Programming	42	4	21	21
		Programming for Mobile Terminals	21	2	0	21
	AI: Fundamentals and techniques	Multimodal AI	21	2	21	0
		Problem-Solving	21	1,5	0	21
		Knowledge and reasoning	21	1,5	21	0
		Optimization Techniques I	21	2	21	0
	Networking and Systems	Operating System Administration: Unix	42	3	21	21
	Business Management	Business Environment	21	1,5	21	0
		Financial Analysis	21	1,5	0	21
	AI Project	AI Project	84	8	42	42
	Communication, Culture et Citoyenneté F3	Communication, Culture et Citoyenneté F3	42	3	21	21

Plan d'étude parcours 4IA

	UE	ECUE	Total hours	Credits (ECTS)	P1 (Hours)	P2 (Hours)
Semestre 7	Numerical Methods	Probabilities	42	3	21	21
		Graph Theory	21	2	0	21
		Statistics	42	3	21	21
	Edge Artificial Intelligence	Deep Learning Project	84	6	42	42
		Introduction to robotics	21	2	21	0
	Data management	Big Data and NoSQL Databases	42	3	21	21
		Database administration	21	2	0	21
	Management and Marketing	Digital Marketing	21	1,5	0	21
		Project Management	21	1,5	21	0
	Communication, Culture et Citoyenneté F4	Communication, Culture et Citoyenneté F4	42	3	21	21
Semestre 8	Internship	Stage d'immersion en entreprise		3		
	Advanced Deep Learning	Natural Language Processing	21	2	21	0
		Natural Language Understanding	21	2	0	21
		Time series	21	2	0	21
	Immersive Reality	XR	42	3	0	42
		Unity	21	2	21	0
	Markov Decision Process and RL	Markov Decision Process	21	2	21	0
		Reinforcement Learning	21	2	0	21
	Advanced AI Project	Advanced AI Project	84	8	42	42
	AI on the cloud	Cloud fundamentals	21	1	21	0
		Service and network administration	21	2	21	0
	Management et pilotage des projets	Innovation and entrepreneurship	21	1	0	21
	English For Engineers A4	English For Engineers A4	42	3	21	21

Plan d'étude parcours 5IA

	UE	ECUE	Total hours	Credits (ECTS)	P1 (Hours)	P2 (Hours)
Semestre 9	Advanced ML Engineering & Operations	DevOps	21	3	21	0
		MLOps	21	3	21	0
		Automation for Business	14	1	14	0
	Humanities and Social science	Cognitive Psychology	21	2	12	9
		Ethics, AI and society	21	3	15	6
		Philosophy of science	12	1	0	12
	Emerging trends in AI	Quantum computing & AI	21	3	21	0
		Article Review	12	1	12	0
		Explainable AI (XAI)	6	1	0	6
	Preparing for the Professional World	Sustainable Development	15	1	15	0
		Personal competencies A	15	2	15	0
		Personal competencies F	15	2	0	15
	Séminaires	Blockchain	15	1	15	0
		AWS Fundamentals	21	1	15	0
Semestre 10	Projet de fin d'études (PFE)	Projet de fin d'études (PFE)		5		